



SYDNEY BOYS HIGH
MOORE PARK, SURRY HILLS

JUNE 2007
TASK #2
YEAR 12

Mathematics Extension 2

General Instructions:

- Reading time—5 minutes.
- Working time—2 hours.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may NOT be awarded for messy or badly arranged work.
- Start each NEW section in a separate answer booklet.
- Hand in your answer booklets in 3 sections:
Section A(Questions 1 and 2),
Section B(Questions 3 and 4),
Section C(Questions 5 and 6),

Total marks—90 Marks

- Attempt questions 1–6.
- All questions are of equal value.

Examiner: Mr D. Hespe

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

Section A

Marks

Question 1 (15 marks)

(a) Find $\int x \sec^2(1 - x^2) dx$. 2

(b) Find $\int \frac{x^2 - 4x + 5}{x - 2} dx$. 2

(c) (i) Find the values of constants A , B , and C if 2

$$\frac{A}{x+1} + \frac{Bx+C}{x^2+1} = \frac{4x^2-2x}{(x+1)(x^2+1)}.$$

(ii) Hence or otherwise find $\int \frac{4x^2-2x}{(x+1)(x^2+1)} dx$. 3

(d) Given that $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin x + \cos x} dx$, 3

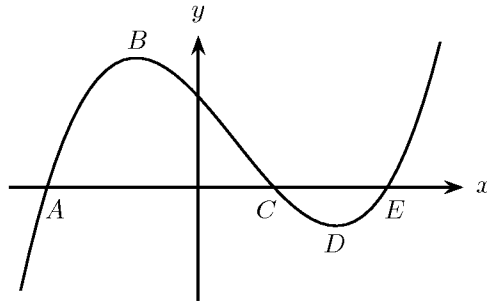
evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x} dx$.

(e) Use the method of integration by parts to evaluate $\int_0^{\ln 2} xe^{-2x} dx$. 3

Question 2 (15 marks)

- (a) A sketch of
- $y = f'(x)$
- is shown below:

3



Draw a neat sketch of a *possible* representation of $f(x)$. Mark on your diagram the points A' , B' , C' , D' , and E' which are related to the points A , B , C , D , and E .

- (b) (i) Draw a neat sketch of $f(x) = (x + 1)(2 - x)(x - 4)$ showing the intercepts on the axes.

2

- (ii) Without using calculus, use your graph of $f(x) = (x + 1)(2 - x)(x - 4)$ to sketch, on separate axes, the graphs of:

(α) $f(x) = |(x + 1)(2 - x)(x - 4)|$

1

(β) $f(x) = \frac{1}{(x + 1)(2 - x)(x - 4)}$

2

(γ) $f(x) = \sqrt{(x + 1)(2 - x)(x - 4)}$

2

- (c) Draw a neat sketch of $y = \sin x \sin 2x$ for $-2\pi \leq x \leq 2\pi$ by considering

5

- (i) the zeroes of the function,
- (ii) the sign of the function in different sections of its domain,
- (iii) whether the function is odd or even.

Section B

(Use a separate writing booklet.)

Marks

Question 3 (15 marks)

(a) Given that $z = -2 + 2i$ and $w = -1 - \sqrt{3}i$,

(i) find

(α) $z - w$,

1

(β) $\Re(z^2)$,

1

(γ) $|w|$,

1

(δ) $z\bar{w}$.

1

(ii) Write z , $\bar{w}$, and $\frac{z}{\bar{w}}$ in modulus and argument form.

3

(iii) Hence determine the number λ which satisfies

2

$$\lambda \left(\frac{z}{\bar{w}} \right) = \left(\frac{z}{\bar{w}} \right)^{25}$$

.

(b) If the argument of the complex number $(z - 1)/(z + 1)$ is $\frac{1}{4}\pi$ show that z lies upon a fixed circle whose centre is at the point which represents i .

3

(c) Find a complex number z whose argument is $\frac{1}{6}\pi$ such that

3

$$|z - \sqrt{3} - i| = |z - 2\sqrt{3} - 2i|$$

Question 4 (15 marks)

(a) Factorise $z^4 + z^2 - 6$ over

(i) $\mathbb{R}$

1

(ii) $\mathbb{C}$

1

(b) If α, β, γ are roots of the equation $2x^3 - 4x^2 - 6x + 5 = 0$, find the cubic equation with roots $\alpha - 1, \beta - 1, \gamma - 1$.

2

(c) It is suspected that $12x^3 - 4x^2 - 5x + 2 = 0$ has a repeated root. Use this to find all the solutions.

4

(d) (i) The roots of the quadratic polynomial $x^2 + px + q = 0$ (where $q \neq 0$) are α and β , and one root of the quadratic equation $x^2 + p'x + q = 0$ is $k\alpha$. Show that the other root of this equation is $\frac{\beta}{k}$.

2

(ii) Assuming that $k^2 \neq 1$, write down for each equation an expression for the sum of its roots and hence find α and β in terms of p, p' , and k .

2

(iii) Deduce that $k(kp - p')(kp' - p) = (k^2 - 1)^2 q$.

1

(iv) For any given values of p, p' , and q , show that the sum of the four possible values of k is $\frac{pp'}{q}$.

2

Section C

(Use a separate writing booklet.)

Marks

Question 5 (15 marks)

- (a) If no two boys are to sit together, in how many ways can six girls and three boys be arranged
- (i) in a row, 2
- (ii) around a circular table? 1
- (b) A pair of integers is selected at random from the set of positive integers $1, 2, 3, \dots, n$.
- (i) In how many ways may the selection be made? 1
- (ii) If n is an odd integer, show that the probability that both integers in the pair selected will be odd is $\frac{n+1}{4n}$. 2
- (iii) If n is even, find the probability that both integers in the pair are odd. 2
- (c) Establish a reduction formula for $\mathbf{I}_n = \int \sec^n x \, dx$. 2
- (d) (i) Any integer n can be expressed in one of the forms $4p, 4p+1, 4p+2, 4p+3$, where p is an integer. Find, for each of these forms, the value of the expression $(1+i^n)(1+i^{2n})$ where $i^2 = -1$. 4
- (ii) Hence write down a function of n which has the value 366 when n is a multiple of 4 and 365 for all other integral values of n 1

Question 6 (15 marks)

(a) $\int_0^2 \sqrt{4+x^2} dx$ 2

[NOTE: you may find the result from Q5(c) useful.]

(b) (i) Determine how the number of stationary points on the curve $y = e^{-x^2}(x^2+c)$ depends on the value of the real constant c . 2

(ii) Find the coördinates of the points of inflexion on the curve for which $c = 2$. 3

(iii) Sketch the curve for which $c = 2$. 2

(c) A rock of mass 5 kg is projected vertically upwards into the air from the ground with initial speed of 12 ms^{-1} . The rock is subject to air resistance of $\frac{v^2}{2}$ Newtons in the opposite direction to its velocity, $v \text{ ms}^{-1}$. The rock is also subject to a downward gravitational force of 50 Newtons.

(i) Find the greatest height reached by the rock. 2

(ii) Find the time taken to reach the highest point. 2

(iii) Find how fast the rock is travelling when it hits the ground. 2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

QUESTION 1

$$(a) \int x \sec^2(1-x^2) dx \quad 2$$

$$= -\frac{1}{2} \tan(1-x^2) + C$$

$$(b) \int \frac{x^2 - 4x + 5}{x-2} dx \quad 2$$

$$= \int \left(x - 2 + \frac{1}{x-2} \right) dx \quad ①$$

$$= x^2 - 2x + \frac{1}{2} \ln|x-2| + C$$

$$(c) \quad (i) \quad 4x^2 - 2x = A(x^2+1) + (Bx+C)(x+1) \quad 2$$

$$A=3 \quad B=1 \quad C=-3$$

$$\therefore I = \int \left(\frac{3}{x+1} + \frac{x-3}{x^2+1} \right) dx \quad 3$$

$$= 3 \ln|x+1| + \frac{1}{2} \ln(x^2+1) - 3 \tan^{-1} x + C$$

$$(d) \quad I = \frac{1}{2} \int \frac{\sin^3 x + \cos^3 x}{\sin x + \cos x} dx \quad 3$$

$$= \frac{1}{2} \int \frac{\sin^2 x + \cos^2 x - \sin x \cos x}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int \left(1 - \frac{1}{2} \sin 2x \right) dx \quad ②$$

$$= \frac{1}{2} \left(x - \frac{1}{2} \cdot -\cos 2x \cdot \frac{1}{2} \right)$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{s^3}{s^2+c} dx = \left[\frac{x}{2} + \frac{1}{8} \cos 2x \right]_0^{\frac{\pi}{2}} = \frac{\pi-1}{4}$$

$$(e) \int_0^{\ln 2} x e^{-2x} dx$$

$$= \int_0^{\ln 2} x \cdot \frac{d}{dx} \left(\frac{1}{2} e^{-2x} \right) dx \quad 3$$

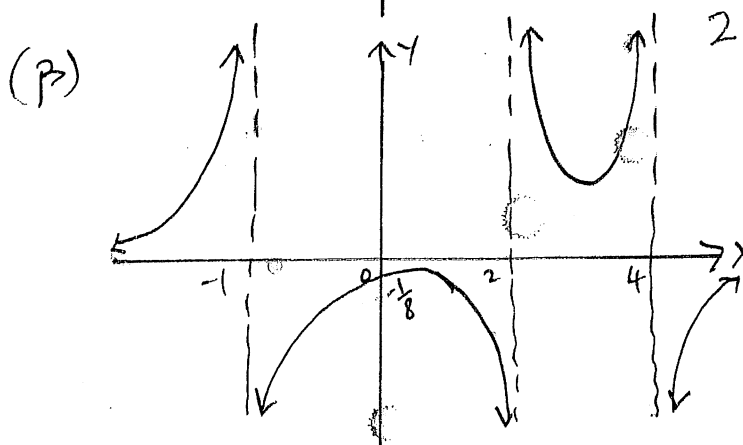
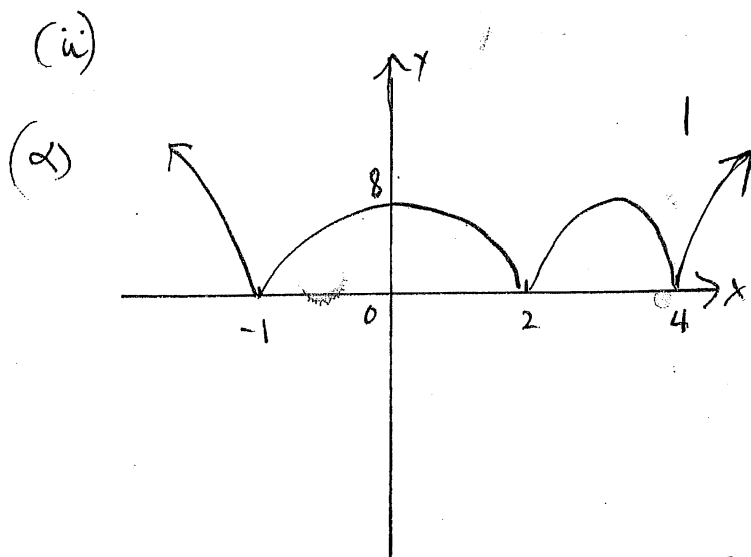
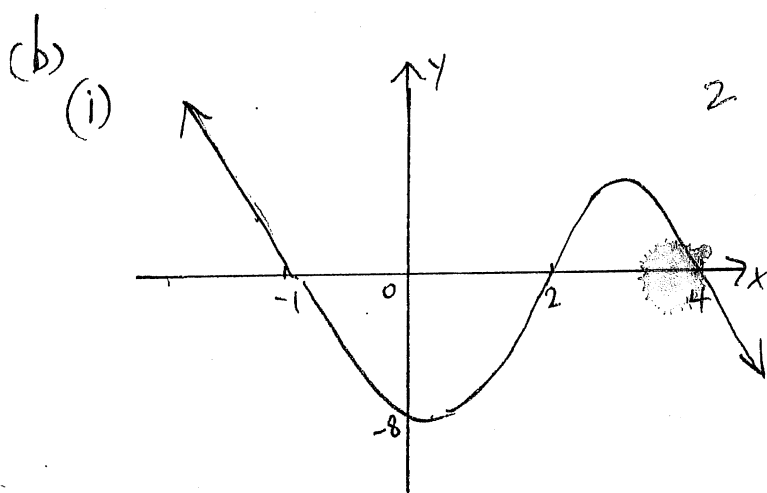
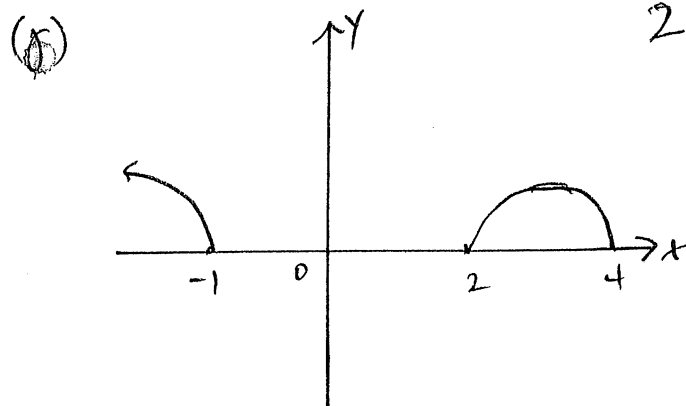
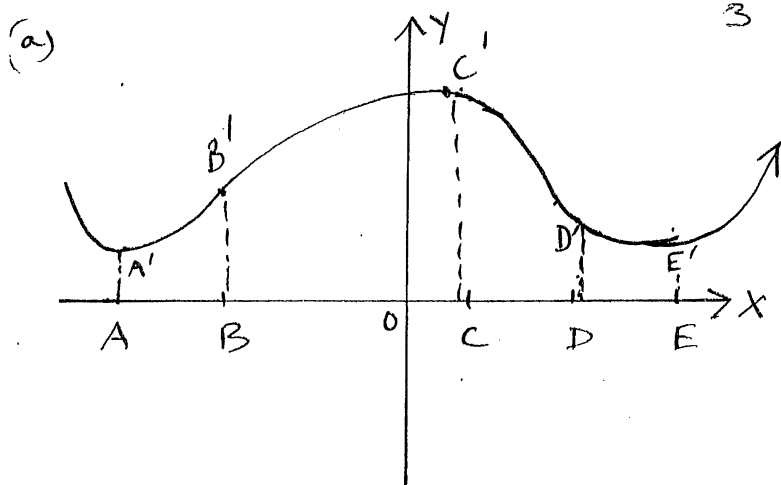
$$= \left[-\frac{1}{2} x \cdot e^{-2x} \right]_0^{\ln 2} - \int_0^{\ln 2} -\frac{1}{2} \cdot e^{-2x} \cdot 1 dx \quad ④$$

$$= \left[-\frac{\ln 2}{2} e^{-2 \ln 2} \right] + \left[-\frac{1}{4} \right] \left[e^{-2 \ln 2} - e^0 \right]$$

$$= -\frac{\ln 2}{2} \cdot 2^{-2} + \left(-\frac{1}{4} \right) (2^{-2} - 1)$$

$$= -\frac{1}{8} \ln 2 + \frac{3}{16}$$

QUESTION 2

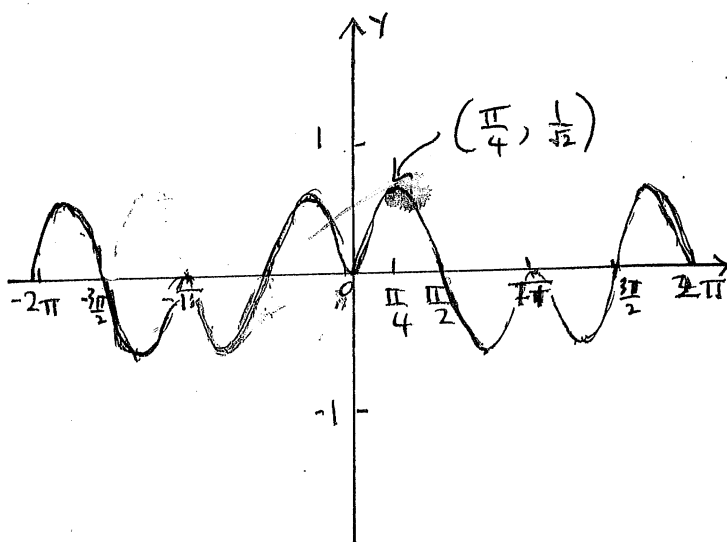


(c) $\sin x \sin 2x = 0 \Rightarrow 5$

$\sin x = 0$
 $x = -2\pi, -\pi, 0, \pi, 2\pi$

OR

$\sin 2x = 0$
 i.e. $2x = -4\pi, -3\pi, -2\pi, -\pi, 0, \pi, 2\pi, 3\pi, 4\pi$
 $\therefore x = -2\pi, -\frac{3\pi}{2}, -\pi, -\frac{\pi}{2}, 0, \pi, \frac{3\pi}{2}, 2\pi$



$f(x) = \sin x \sin 2x$

$f(-x) = \sin(-x) \cdot \sin(-2x)$

$= -\sin x \cdot -\sin 2x$

$= \sin x \cdot \sin 2x$

$= f(x) \therefore f(x) \text{ even}$

Section B Question 3

(a) $z = -2 + 2i$ $w = -1 - \sqrt{3}i$

(i) (a) $z - w = (-2 + 2i) - (-1 - \sqrt{3}i)$
 $= -1 + (2 + \sqrt{3})i$ [1]

(b) $\operatorname{Re}(z^2) = \operatorname{Re}(-2 + 2i)^2$
 $= \operatorname{Re}(-8i)$
 $= 0$ [1]

(c) $|w| = \sqrt{1^2 + \sqrt{3}^2}$
 $= 2$ [1]

(d) $z\bar{w} = (-2 + 2i)(-1 + \sqrt{3}i)$
 $= 2 - 2\sqrt{3}i - 2i - 2\sqrt{3}$
 $= (2 - 2\sqrt{3}) - (2 + 2\sqrt{3})i$ [1]

(ii) $z = -2 + 2i$ $\arg z = \frac{3\pi}{4}$
 $|z| = 2\sqrt{2}$

$\therefore z = 2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}$ [1]

$\bar{w} = -1 + \sqrt{3}i$ $\arg \bar{w} = \frac{2\pi}{3}$
 $|\bar{w}| = 2$

$\therefore \bar{w} = 2 \operatorname{cis} \frac{2\pi}{3}$ [1]

$\frac{z}{\bar{w}} = \frac{2\sqrt{2} \operatorname{cis} \frac{3\pi}{4}}{2 \operatorname{cis} (\frac{2\pi}{3})}$
 $= \sqrt{2} \operatorname{cis} (\frac{3\pi}{4} - \frac{2\pi}{3})$
 $= \sqrt{2} \operatorname{cis} \frac{\pi}{12}$ [1]

(iii) $\lambda \left(\frac{z}{\bar{w}} \right) = \left(\frac{z}{\bar{w}} \right)^{25}$
 $\therefore \lambda = \left(\frac{z}{\bar{w}} \right)^{24}$

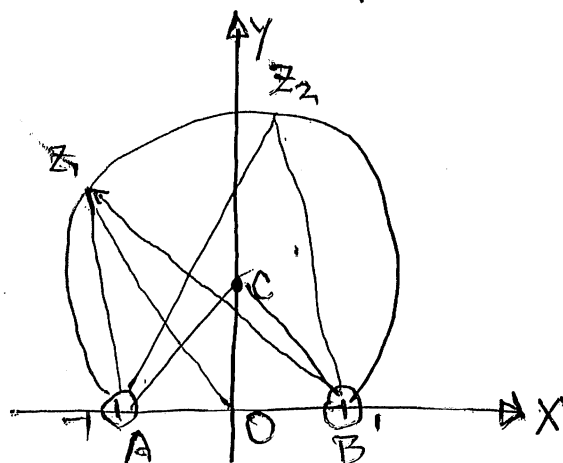
$= (\sqrt{2})^{24} \operatorname{cis} \left(\frac{\pi}{12} \times 24 \right)$

$= 4096 \operatorname{cis} 2\pi$

$= 4096$

[2]

(b) $\arg \left(\frac{z-1}{z+1} \right) = \frac{\pi}{4}$



Let Z_1, Z_2 be points representing two values of z . Now $\angle AZ_1B = \angle AZ_2B = \frac{\pi}{4}$

$\therefore$ By the theorem, all z lie on a circle.

Let C be the centre of the circle.

$\therefore \angle ACB = \frac{\pi}{2}$ (angle at the centre twice angle at circumference).

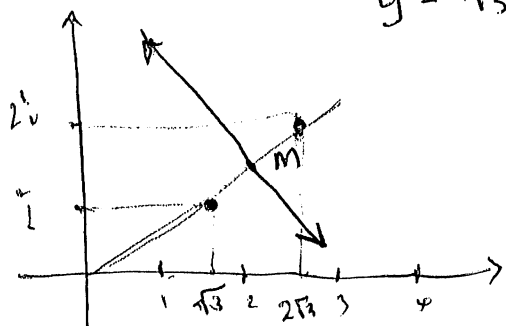
$\therefore$ By symmetry and similarity, $OC = 1$

$\therefore$ Centre is i . [3]

$$(c) \arg z = \frac{\pi}{6}$$

$$|z - (\sqrt{3} + i)| = |z - 2(\sqrt{3} + i)|$$

This is the perp. bisector of the interval joining $\sqrt{3} + i$ and $2\sqrt{3} + 2i$
 $y = -\sqrt{3}x + 3$



$$\text{Midpoint } M \text{ is } \frac{3\sqrt{3}}{2} + \frac{3i}{2}$$

We seek z lying on this line with $\arg z = \frac{\pi}{6}$.

$$\text{But } \arg(M) = \frac{\pi}{6}$$

$$\therefore z = M$$

$$= \frac{3}{2}(\sqrt{3} + i)$$

$$= 3 \cos \frac{\pi}{6}$$

Question 4

$$(a) z^4 + z^2 - 6$$

$$= (z^2 + 3)(z^2 - 2)$$

$$(i) = (z^2 + 3)(z + \sqrt{3}i)(z - \sqrt{3}i)$$

$$(ii) = (z + \sqrt{3}i)(z - \sqrt{3}i)(z + \sqrt{2})(z - \sqrt{2})$$

$$(b) 2x^3 - 4x^2 - 6x + 5 = 0 \text{ has}$$

roots α, β, γ .

$$\text{Let } y = x - 1$$

$$\therefore x = y + 1$$

$$\text{Thus } 2(y+1)^3 - 4(y+1)^2 - 6(y+1) + 5 = 0$$

has roots $\alpha - 1, \beta - 1, \gamma - 1$

Simplifying:

$$2(y^3 + 3y^2 + 3y + 1) - 4(y^2 + 2y + 1)$$

$$- 6y - 6 + 5 = 0$$

$$2y^3 + 6y^2 + 6y + 2 - 4y^2 - 8y - 4$$

$$- 6y - 1 = 0$$

$$2y^3 + 2y^2 - 8y - 3 = 0$$

Replace y with x :

$$2x^3 + 2x^2 - 8x - 3 = 0$$

has real roots.

$$(c) 12x^3 - 4x^2 - 5x + 2 = 0 \quad \text{--- (1)}$$

Derived eq'n:

$$36x^2 - 8x - 5 = 0 \quad \text{--- (2)}$$

$$x = \frac{8 \pm \sqrt{64 + 4 \times 180}}{72}$$

$$= \frac{1}{2}, -\frac{5}{18}$$

Test $x = \frac{1}{2}$ in Eq'n (1)

$$12\left(\frac{1}{2}\right)^3 - 4\left(\frac{1}{2}\right)^2 - 5\left(\frac{1}{2}\right) + 2 = 0$$

$\therefore x = \frac{1}{2}$ is a repeated root. (1)

$\therefore (x - \frac{1}{2})^2$ is a factor of (1).

$$x^2 - 1 + \frac{1}{4} \overline{12x^3 - 4x^2 - 5x + 2}$$

$$12x^3 - 12x^2 + 3x$$

$$8x^2 - 8x + 2$$

$$8x^2 - 8x + 2$$

0

$\therefore$ Roots are $\frac{1}{2}, \frac{1}{2}, -\frac{2}{3}$

Q4

$$(d) x^2 + px + q = 0 \quad (\text{Roots } \alpha, \beta) \quad \text{--- (1)}$$

$$x^2 + p'x + q = 0 \quad (\text{Roots } k\alpha, \gamma \text{ say}) \quad \text{--- (2)}$$

$$(i) \text{ From (1) } \alpha\beta = q$$

$$\text{From (2) } k\alpha\gamma = q$$

$$\therefore \alpha\beta = k\alpha\gamma$$

$$\therefore \gamma = \frac{\beta}{k}$$

$$(ii) \text{ For (1) } \alpha + \beta = -p$$

$$\text{For (2) } k\alpha + \frac{\beta}{k} = -p'$$

$$\text{Substitute } \beta = -p - \alpha$$

$$\therefore k\alpha - \frac{p + \alpha}{k} = -p'$$

$$k^2\alpha - (p + \alpha) = -kp'$$

$$k^2\alpha - \alpha = p - kp'$$

$$\alpha(k^2 - 1) = p - kp'$$

$$\alpha = \frac{p - kp'}{k^2 - 1}$$

$$\text{Now } \beta = -p - \alpha$$

$$= -p - \left(\frac{p - kp'}{k^2 - 1} \right)$$

$$= \frac{-p(k^2 - 1) - (p - kp')}{k^2 - 1}$$

$$\therefore \beta = \frac{kp' - k^2p}{k^2 - 1}$$

$$(iii) \text{ Now } \alpha\beta = q$$

$$\therefore \frac{p - kp'}{k^2 - 1} \cdot \frac{k(p' - kp)}{k^2 - 1} = q$$

$$\therefore (p - kp')k(p' - kp) = (k^2 - 1)^2 q$$

$$\therefore k(kp - p')(kp' - p) = (k^2 - 1)^2 q$$

QED [1]

(iv) Expanding both sides and collecting terms on the left gives the quartic eq'n in k :

$$qk^4 - pp'k^3 + \dots = 0$$

$\therefore$ By sum of roots

$$\text{Sum of all } k's = \frac{pp'}{q}$$

QED

Q5(a) (i)

x x x x x x

Arrangements of girls = $6!$

Choose 3 from 7 locations for boys = 7P_3

$$\therefore \text{No of arrangements} = 6! \cdot {}^7P_3$$

$$= 151200$$

2

(ii)

(*)

x
x x
x x

Arrangements of girls = $5!$

Choose 3 from 6 locations for boys = 6P_3

$$\therefore \text{No of arrangements} = 5! \cdot {}^6P_3$$

$$= 14400$$

1

(b) (i) ${}^nC_2 = \frac{n(n-1)}{2}$

(ii) $P(\text{odd, odd}) = \frac{\cancel{n+1}}{n} \cdot \frac{\frac{n-1}{2}}{n-1}$

$$= \frac{n+1}{2n} \cdot \frac{1}{2}$$

$$= \frac{n+1}{4n}$$

$$\frac{\frac{n}{2}-1}{2n-2}$$

$$= \frac{n-2}{4n-4}$$

2

(iii) $P(\text{odd, odd}) = \frac{1}{2} \cdot \frac{\frac{n-2}{2}}{n-1}$

$$= \frac{1}{4} \cdot \frac{n-2}{n-1}$$

$$= \frac{n-2}{4n-4}$$

$$\frac{{}^mC_2}{{}^{2m}C_2}$$

$$= \frac{\frac{m(m-1)}{2}}{\frac{2m(2m-1)}{2}}$$

$$= \frac{m-1}{2m}$$

2

$$(c) \quad I_n = \int \sec^n x \, dx$$

$$= \int \sec^{n-2} x \cdot \sec^2 x \, dx$$

$$= \sec^{n-2} x \cdot \tan x$$

$$u = \sec^{n-2} x \quad v = \tan x$$

$$- \int (n-2) \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x \, dx \quad u' = \sec^{n-2} x$$

$$(n-2) \sec^{n-3} x \cdot \sec x \tan x$$

$$= \sec^{n-2} x \tan x$$

$$- (n-2) \int \sec^{n-2} x \cdot \tan^2 x \, dx$$

$$= \sec^{n-2} x \tan x$$

$$- (n-2) \int \sec^{n-2} x \cdot (\sec^2 x - 1) \, dx$$

$$= \sec^{n-2} x \tan x - (n-2) \left\{ \int \sec^n x \, dx - \int \sec^{n-2} x \, dx \right\}$$

$$\therefore I_n = \sec^{n-2} x \tan x - (n-2) \{ I_n - I_{n-2} \}$$

$$\therefore I_n + (n-2) I_n = \sec^{n-2} x \tan x + (n-2) I_{n-2}$$

$$\therefore I_n (n-1) = \sec^{n-2} x \tan x + (n-2) I_{n-2}$$

$$\therefore I_n = \frac{1}{n-1} \sec^{n-2} x \tan x + \frac{n-2}{n-1} I_{n-2}$$

2

(d) (i) If $n = 4p$

$$\begin{aligned} & (1+i^n)(1+i^{2n}) \\ &= (1+i^{4p})(1+i^{8p}) \\ &= 2 \cdot 2 \\ &= 4 \end{aligned}$$

If $n = 4p+1$

$$\begin{aligned} & (1+i^n)(1+i^{2n}) \\ &= (1+i^{4p+1})(1+i^{8p+2}) \\ &= (1+i)(1+i^2) \\ &= (1+i)(1-i) \\ &= 1 \end{aligned}$$

If $n = 4p+2$

$$\begin{aligned} & (1+i^n)(1+i^{2n}) \\ &= (1+i^{4p+2})(1+i^{8p+4}) \\ &= (1+i^2)(1+i^4) \\ &= 0 \end{aligned}$$

If $n = 4p+3$

$$\begin{aligned} & (1+i^{4p+3})(1+i^{8p+6}) \\ &= (1-i)(1-i) \\ &= 0 \end{aligned}$$

(ii) Function is $f(n) = 365 + \frac{(1+i^n)(1+i^{2n})}{4}$

4

1

$$Q6 (a) \int_0^2 \sqrt{4+x^2} dx$$

$$\text{let } x = 2 \tan \theta$$

$$\therefore dx = 2 \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \sqrt{4+4\tan^2 \theta} \cdot 2 \sec^2 \theta d\theta$$

$$\text{If } x=0, \theta=0$$

$$= \int_0^{\pi/4} 2 \sec \theta \cdot 2 \sec^2 \theta d\theta$$

$$\text{If } x=2, \theta = \frac{\pi}{4}$$

$$= 4 \int_0^{\pi/4} \sec^3 \theta d\theta$$

$$= 4 \left\{ \left[\frac{1}{2} \sec \theta \tan \theta \right]_0^{\pi/4} + \frac{1}{2} \int_0^{\pi/4} \sec \theta d\theta \right\}$$

$$= 4 \left\{ \left[\frac{1}{2} \sec \frac{\pi}{4} \cdot \tan \frac{\pi}{4} \right] - \left[\frac{1}{2} \sec 0 \tan 0 \right] + \frac{1}{2} \left[\ln (\sec \theta + \tan \theta) \right]_0^{\pi/4} \right\}$$

$$= 4 \left\{ \frac{1}{2} \cdot \sqrt{2} \cdot 1 - 0 + \frac{1}{2} \left[\ln (\sec \frac{\pi}{4} + \tan \frac{\pi}{4}) \right] - \frac{1}{2} \ln (\sec 0 + \tan 0) \right\}$$

$$= 2\sqrt{2} + \frac{1}{2} \ln (\sqrt{2} + 1) - \frac{1}{2} \ln 1$$

$$= 2\sqrt{2} + \frac{1}{2} \ln (\sqrt{2} + 1)$$

2

$$(b) (i) \quad y = e^{-x^2} (x^2 + c)$$

$$\begin{aligned} y' &= (x^2 + c) \cdot e^{-x^2} \cdot -2x + e^{-x^2} \cdot 2x \\ &= 2x e^{-x^2} (-x^2 + c + 1) \end{aligned}$$

$$y' = 0 \quad \text{if } x = 0 \quad \text{or } x^2 + c + 1 = 0.$$

$$\therefore x^2 = \cancel{1-c} \quad 1-c$$

$$\therefore \cancel{1-c} \geq 0$$

$$\therefore c \leq 1, \text{ for zeros.}$$

$$\text{If } c = 1 \quad x = 0$$

If $c < 1$ the curve has 3 stationary points.

otherwise the curve has 1 stationary point. 2

$$(ii) \text{ If } c = 2$$

$$\begin{aligned} y' &= -2x e^{-x^2} (x^2 + 1) \\ &= e^{-x^2} (-2x^3 - 2x) \end{aligned}$$

$$\begin{aligned} y'' &= e^{-x^2} \cdot (-6x^2 - 2) + \frac{d}{dx} (-2x) (x^2 + 1) \cdot e^{-x^2} - 2x \\ &= e^{-x^2} (-6x^2 - 2 + 4x^2(x^2 + 1)) \end{aligned}$$

$$= e^{-x^2} (-6x^2 - 2 + 4x^4 + 4x^2)$$

$$= e^{-x^2} (4x^4 - 2x^2 - 2)$$

$$= 2e^{-x^2} (2x^4 - x^2 - 1)$$

$$= 2e^{-x^2} (2x^2 + 1)(x^2 - 1)$$

$$\therefore x^2 = -\frac{1}{2} \quad \text{or } x^2 = 1$$

$$\therefore x = \pm 1$$

∴ Potential points of inflection are

$$(1, \frac{3}{2}) \text{ or } (-1, \frac{3}{2})$$

$$\text{If } x = 0.5 \quad y'' = +ve \times +ve \times -ve < 0$$

$$\text{If } x = 1.5 \quad y'' = +ve \times +ve \times +ve \geq 0$$

∴ Chge of concavity.

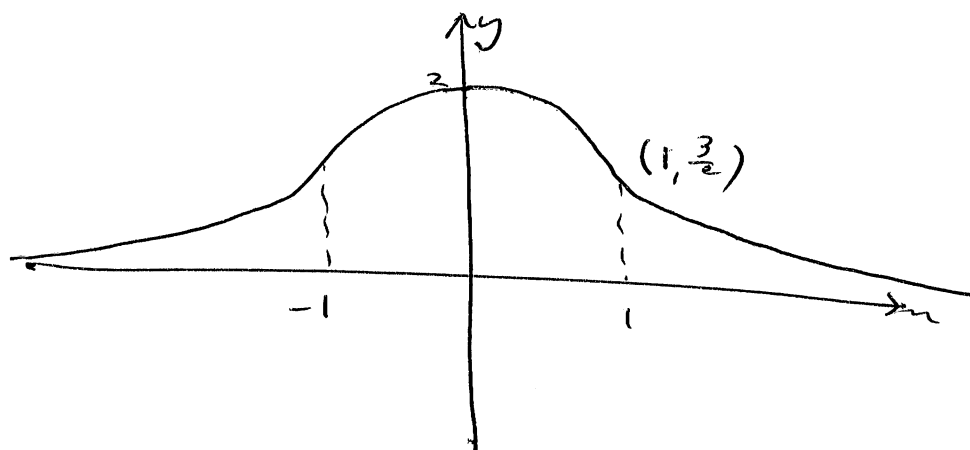
$(1, \frac{3}{2})$ is a point of inflection.

The function is an even function

∴ $(-1, \frac{3}{2})$ is also a point of inflection

(iii) When $x=0$ $y=2$

$$\text{As } x \rightarrow \infty \quad y \rightarrow 0$$



(v) (i)

$$\ddot{x} = -\left(10 + \frac{v^2}{10}\right)$$

↓
so ↓ ↓ $\frac{v^2}{2}$

$$v \frac{dv}{dx} = -\frac{100 + v^2}{10}$$

~~$$\int \frac{v dv}{100 + v^2} = -\int \frac{dx}{10}$$~~

$$\int \frac{v dv}{100 + v^2} = -\int \frac{dx}{10}$$

$$\frac{1}{2} \ln(100 + v^2) = -\frac{1}{10} x + c$$

when $x=0$, $v=12$

$$\therefore \frac{1}{2} \ln 244 = c$$

$$\therefore -\frac{1}{10} x = \frac{1}{2} \ln(100 + v^2) - \frac{1}{2} \ln 244$$

$$\therefore \frac{1}{10} x = \frac{1}{2} \left\{ \ln(244) - \ln(100 + v^2) \right\}$$

$$\therefore \frac{1}{5} x = \ln \frac{244}{100 + v^2}$$

Max ht reached when $v=0$

$$\therefore \frac{1}{5} x = \ln \frac{244}{100}$$

$$\therefore x = 5 \ln \frac{244}{100}$$

$$= 4.4899 \dots \text{ m}$$

2

$$(ii) \quad \frac{dr}{dt} = -\left(10 + \frac{r^2}{10}\right)$$

$$\therefore \frac{dr}{dt} = -\frac{100 + r^2}{10}$$

$$\therefore \int \frac{dr}{100 + r^2} = \int -\frac{1}{10} dt.$$

$$\therefore \frac{1}{10} \tan^{-1} \frac{r}{10} = -\frac{1}{10} t + c.$$

When $t=0, r=12$

$$\frac{1}{10} \tan^{-1} \frac{12}{10} = c.$$

$$\therefore \frac{1}{10} \tan^{-1} \frac{r}{10} = -\frac{1}{10} t + \frac{1}{10} \tan^{-1} \frac{12}{10}.$$

$$\therefore \tan^{-1} \frac{r}{10} = -t + \tan^{-1} \frac{12}{10}.$$

$$\therefore t = \tan^{-1} \frac{12}{10} - \tan^{-1} \frac{r}{10}.$$

If $r=0, t = \tan^{-1} \frac{12}{10}$
 $= 0.8760 \dots s$

2

~~scribble~~

(iii)

$$\uparrow \frac{v^2}{2}$$

$$\ddot{x} = 10 - \frac{v^2}{10}$$

$$\downarrow 10$$

$$\frac{v dv}{dx} = \frac{100 - v^2}{10}$$

$$\therefore \int \frac{v dv}{100 - v^2} = \int \frac{1}{10} dx$$

$$-\frac{1}{2} \ln(100 - v^2) = \frac{1}{10} x + c$$

$$\text{When } x=0, v=0: -\frac{1}{2} \ln 100 = c$$

$$\therefore -\frac{1}{2} \ln(100 - v^2) = \frac{1}{10} x - \frac{1}{2} \ln 100$$

$$\frac{1}{10} x = \frac{1}{2} (\ln 100 - \ln(100 - v^2))$$

$$x = 5 \left(\ln \frac{100}{100 - v^2} \right)$$

$$\text{If } x = 5 \ln \frac{244}{100} :$$

$$5 \ln \frac{244}{100} = 5 \ln \frac{100}{100 - v^2}$$

$$\therefore \frac{244}{100} = \frac{100}{100 - v^2}$$

$$\therefore 100 - v^2 = \frac{10000}{244}$$

$$v^2 = 100 - \frac{10000}{244}$$

$$v^2 = \frac{24400 - 10000}{244}$$

$$= \frac{14400}{244}$$

$$\therefore v = \frac{120}{\sqrt{244}}$$

$$= 27.6222 \text{ m/s}$$

2